Kanamori Note

Anderson Steckler

1 Hamiltonian

Our single layer Hamiltonian is given by the following:

$$H = H_0 + H_{SOC} + H_{Kanamori} \tag{1}$$

the tight binding part of the Hamiltonian is given by (yz, xz, xy) orbital basis

$$\begin{aligned}
e_{yz} &= -2t_1(\cos k_x + \cos k_y) - 4t_2 \cos k_x \cos k_y && (2)\\
e_{xz} &= -2t_1 \cos k_x - 2t_\delta \cos k_y && (3)\\
e_{xy} &= -2t_\delta \cos k_x - 2t_1 \cos k_y && (4)\\
& && (5)
\end{aligned}$$

$$H_0(k) = \begin{pmatrix} e_{yz} & 0 & 0 \\ 0 & e_{xz} & 0 \\ 0 & 0 & e_{xy} + \delta_{CF} \end{pmatrix} \tag{6}$$

where Δ_{CF} is the crystal field splitting. For the hoppings t_1 represents the in plane $V_{dd\pi}$ overlap. t_2 represents the diagonal nearest neighbor hopping. t_δ represents a weak hopping which comes from the $V_{dd\delta}$ overlap. The spin orbit coupling term is given by

$$H_{SOC} = \lambda \boldsymbol{L} \cdot \boldsymbol{S} \tag{7}$$

where λ is the strength of the spin orbit coupling. The Kanamori interaction term is done in mean-field and derivation is given at the end. The result is this:

$$\begin{aligned}
H^k_{MF} = U &\sum_m \langle n_{m\uparrow}\rangle \hat{n}_{m\downarrow} + \langle n_{m\downarrow}\rangle \hat{n}_{m\uparrow} \qquad (8)\\
&+ U' \sum_{m\neq m'} \langle n_{m\uparrow}\rangle \hat{n}_{m'\downarrow} + \langle n_{m'\downarrow}\rangle \hat{n}_{m\uparrow}\\
&+ (U' - J) \sum_{m<m',\sigma} \langle n_{m\sigma}\rangle \hat{n}_{m'\sigma} + \langle n_{m'\sigma}\rangle \hat{n}_{m\sigma}\\
&+ J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle d^\dagger_{m'\downarrow} d_{m\downarrow} + \langle d^\dagger_{m'\downarrow} d_{m\downarrow}\rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m\downarrow}\rangle d^\dagger_{m'\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m'\downarrow} d_{m'\uparrow}\rangle d^\dagger_{m\uparrow} d_{m\downarrow} \right]\\
&+ J \sum_{m\neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow}\rangle d^\dagger_{m\downarrow} d_{m'\downarrow} + \langle d^\dagger_{m\downarrow} d_{m'\downarrow}\rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m'\downarrow}\rangle d^\dagger_{m\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m\downarrow} d_{m'\uparrow}\rangle d^\dagger_{m\uparrow} d_{m'\downarrow} \right]
\end{aligned}$$

2 Bilayer Hamiltonian

The bilayer Hamiltonian is given by

$$H = \begin{pmatrix} H_0 & H_\perp \\ H_\perp & H_0 \end{pmatrix} \tag{9}$$

With the interlayer hoppings given by

$$H_\perp = \begin{pmatrix} t_\perp & 0 & 0 \\ 0 & t_\perp & 0 \\ 0 & 0 & t_{xy} \end{pmatrix} \tag{10}$$

3 Basis Ordering

Our order is layer- spin - orbital.

Index	Layer	Spin	Orbital
0	L_1	$\uparrow$	d_{yz}
1	L_1	$\uparrow$	d_{xz}
2	L_1	$\uparrow$	d_{xy}
3	L_1	$\downarrow$	d_{yz}
4	L_1	$\downarrow$	d_{xz}
5	L_1	$\downarrow$	d_{xy}
6	L_2	$\uparrow$	d_{yz}
7	L_2	$\uparrow$	d_{xz}
8	L_2	$\uparrow$	d_{xy}
9	L_2	$\downarrow$	d_{yz}
10	L_2	$\downarrow$	d_{xz}
11	L_2	$\downarrow$	d_{xy}

Table 1: Basis state ordering: layer-major $\rightarrow$ spin-major $\rightarrow$ orbital-minor.

4 Self Consistent Loop

Our order parameter is the density matrix

$$\rho_{ab} = \langle c_a^\dagger c_b \rangle = \frac{1}{N_k} \sum_{\mathbf{k},n} f(\varepsilon_{n\mathbf{k}} - \mu)\, v^*_{n\mathbf{k},a}\, v_{n\mathbf{k},b} \tag{11}$$

where a,b run over the 12 states. Wit our parameters we first run the SCF loop with just the hubbard hamiltonian to get a diagonal initial guess for the density matrix. Then we break the symmetry. Doing this in different ways can converge on different results so you would have to take the one of lowest energy. For example I get interesting results comparable to the DFT with this.

$$\delta\rho = \delta \begin{pmatrix} +1 & +1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ +1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & +1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & +1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & +1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & +1 & +1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & +1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$$

Which AI reccomended I need to think about the meaning of each of these and why. We then do the diagonalization and calculate the new density matrix using the convergance criteria as

$$||\rho_{calc} - \rho_{old}|| < \epsilon = 10^{-6} \tag{12}$$

where this is the Frobenius norm. Flatten the matrix and take the norm.

5 Results

5.1 Hunds Rule

We turn off hopping and SOC. With $U = 4\ eV$ and $J = 0.8\ eV$. We also set $t_\perp$ to zero so the layers are decoupled. this follows the first Hund's rule of maxamizing S. But does not maximize L. This is because the

Orbital	Spin Up	Spin Down
d_{xy}	1.0000	0.6667
d_{xz}	1.0000	0.6667
d_{yz}	1.0000	0.6667
Total	5.0000	

Table 2: Orbital occupancies.

guess for ρ_0 was diagonal. Adding some shift (I didn't write down exactly what it was) we get Where now the

Orbital	Spin Up	Spin Down
dxy	1.0000	1.0000
dxz	1.0000	1.0000
dyz	1.0000	0.0000
Total	5.0000	

Table 3: Orbital occupancies.

yz is unoccupied which corresponds to angular of $m = -1$ in the TP equivelence. So now the second rule is satisfied of maximizing L.

One more comment is that I tried some different shift and got that my code did not converge. Again I forgot to write it down which is too bad, but I can try to reproduce it later. I think in general the shift to the diagonal ρ_0 needs to align with the expected result.

5.2 Charge Disportion

I added a different splitting of the xy orbital for each layer, one positive and one negative with slightly less absoulte value. Using the shift to the initial density matrix above (the big 12x12) one I get. If we sum the

	$n_{xz/yz\downarrow}$	$n_{xy\downarrow}$	$n_\downarrow$ (total t2g)
Co1 DFT	0.98	0.32	2.28
Co1 Model	0.78	0.37	1.94
Co2 DFT	0.38	1.00	1.76
Co2 Model	0.62	0.82	2.06

Table 4: Comparison of Sr3Co2O7 DFT (table 2) and model orbital occupancies.

occupations and subtract each layer we get a similar value to the dft 0.13 vs 0.14, so these crystal field values capture this result. Likewise we see in one layer the xy orbital has the higher occupation which is reverse to the other layer. But the ratios are not exactly the same. This could be due to a mean field artifact, due to it following Hund's first rule it wants the occupations of the spin up channel to always be one.

We used the following parameters.

Parameter	Value
t_1	0.3 eV
t_δ	0.05 eV
t_2	0.03 eV
$t_\perp$	0.2 eV
$t_\perp^{xy}$	0.05 eV
λ	0.1 eV
U	4.0 eV
J	0.8 eV
U'	2.4 eV
δ_{cf}^{L1}	+0.23 eV
δ_{cf}^{L2}	-0.16 eV

Table 5: Model parameters.

These were just suggested from AI I need to go through on my own and think about if these are reasonable.

6 Kanamori Hamiltonian

To capture the complete physics we extend out model to the Kanamori Hamiltonian

$$\begin{aligned} H_K = U \sum_m \hat{n}_{m\uparrow}\hat{n}_{m\downarrow} + U' \sum_{m \neq m'} \hat{n}_{m\uparrow}\hat{n}_{m'\downarrow} + (U' - J) \sum_{m<m',\sigma} \hat{n}_{m\sigma}\hat{n}_{m'\sigma} \\ - J \sum_{m \neq m'} d^\dagger_{m\uparrow} d_{m\downarrow} d^\dagger_{m'\downarrow} d_{m'\uparrow} + J \sum_{m \neq m'} d^\dagger_{m\uparrow} d^\dagger_{m\downarrow} d_{m'\downarrow} d_{m'\uparrow} \end{aligned} \tag{13}$$

We have already looked at the MF treatment of the first term above.

6.1 Onsite Repulsion

$$H = U \sum_m \langle n_{m\uparrow} \rangle \hat{n}_{m\downarrow} + \langle n_{m\downarrow} \rangle \hat{n}_{m\uparrow} - \langle n_{m\uparrow} \rangle \langle n_{m\downarrow} \rangle \tag{14}$$

6.2 Interorbital Repulsion

6.2.1 Opposite Spin

$$H = U' \sum_{m \neq m'} \hat{n}_{m\uparrow}\hat{n}_{m'\downarrow} \tag{15}$$

As before each operator is written as

$$\hat{n}_{m\sigma} = \langle n_{m\sigma} \rangle + \delta n_{m\sigma} \tag{16}$$

which allows us to write

$$
\begin{aligned}
H &= U' \sum_{m \neq m'} [\langle n_{m\uparrow} \rangle + \delta n_{m\uparrow}][\langle n_{m'\downarrow} \rangle + \delta n_{m'\downarrow}] \\
&= U' \sum_{m \neq m'} \langle n_{m\uparrow} \rangle \langle n_{m'\downarrow} \rangle + \langle n_{m\uparrow} \rangle \delta n_{m'\downarrow} + \langle n_{m'\downarrow} \rangle \delta n_{m\uparrow}
\end{aligned} \tag{17}
$$

with $\delta n_{m\sigma} = \hat{n}_{m\sigma} - \langle n_{m\sigma} \rangle$

$$
\begin{aligned}
H &= U' \sum_{m \neq m'} \langle n_{m\uparrow} \rangle \langle n_{m'\downarrow} \rangle + \langle n_{m\uparrow} \rangle [\hat{n}_{m'\downarrow} - \langle n_{m'\downarrow} \rangle] + \langle n_{m'\downarrow} \rangle [\hat{n}_{m\uparrow} - \langle n_{m\uparrow} \rangle] \\
&= U' \sum_{m \neq m'} \langle n_{m\uparrow} \rangle \hat{n}_{m'\downarrow} + \langle n_{m'\downarrow} \rangle \hat{n}_{m\uparrow} - \langle n_{m\uparrow} \rangle \langle n_{m'\downarrow} \rangle
\end{aligned} \tag{18}
$$

6.2.2 Same Spin

The next term is

$$
\begin{aligned}
H &= (U' - J) \sum_{m < m', \sigma} \hat{n}_{m\sigma} \hat{n}_{m'\sigma} \\
&= (U' - J) \sum_{m < m', \sigma} [\langle n_{m\sigma} \rangle + \delta n_{m\sigma}][\langle n_{m'\sigma} \rangle + \delta n_{m'\sigma}] \\
&= (U' - J) \sum_{m < m', \sigma} \langle n_{m\sigma} \rangle \langle n_{m'\sigma} \rangle + \langle n_{m\sigma} \rangle \delta n_{m'\sigma} + \langle n_{m'\sigma} \rangle \delta n_{m\sigma}
\end{aligned} \tag{19}
$$

with $\delta n_{m\sigma} = \hat{n}_{m\sigma} - \langle n_{m\sigma} \rangle$

$$
\begin{aligned}
H &= (U' - J) \sum_{m < m', \sigma} \langle n_{m\sigma} \rangle \langle n_{m'\sigma} \rangle + \langle n_{m\sigma} \rangle [\hat{n}_{m'\sigma} - \langle n_{m'\sigma} \rangle] + \langle n_{m'\sigma} \rangle [\hat{n}_{m\sigma} - \langle n_{m\sigma} \rangle] \\
&= (U' - J) \sum_{m < m', \sigma} \langle n_{m\sigma} \rangle \hat{n}_{m'\sigma} + \langle n_{m'\sigma} \rangle \hat{n}_{m\sigma} - \langle n_{m\sigma} \rangle \langle n_{m'\sigma} \rangle
\end{aligned} \tag{20}
$$

6.3 Exchange/Pair-Hopping Term

The terms are

$$
H = -J \sum_{m \neq m'} d^\dagger_{m\uparrow} d_{m\downarrow} d^\dagger_{m'\downarrow} d_{m'\uparrow} + J \sum_{m \neq m'} d^\dagger_{m\uparrow} d^\dagger_{m\downarrow} d_{m'\downarrow} d_{m'\uparrow} \tag{21}
$$

We can normal order and get

$$
H = J \sum_{m \neq m'} \left[d^\dagger_{m\uparrow} d^\dagger_{m'\downarrow} d_{m\downarrow} d_{m'\uparrow} + d^\dagger_{m\uparrow} d^\dagger_{m\downarrow} d_{m'\downarrow} d_{m'\uparrow} \right] \tag{22}
$$

Here we expand with mean field

$$
\begin{aligned}
d^\dagger_1 d^\dagger_2 d_3 d_4 = -\langle d^\dagger_1 d_3 \rangle d^\dagger_2 d_4 + \langle d^\dagger_1 d_4 \rangle d^\dagger_2 d_3 - \langle d^\dagger_2 d_4 \rangle d^\dagger_1 d_3 + \langle d^\dagger_2 d_3 \rangle d^\dagger_1 d_4 \\
+ DC
\end{aligned} \tag{23}
$$

Where DC corrects for double counting and is

$$
DC = \langle d^\dagger_1 d_4 \rangle \langle d^\dagger_2 d_3 \rangle - \langle d^\dagger_1 d_3 \rangle \langle d^\dagger_2 d_4 \rangle \tag{24}
$$

Where we drop the $\langle dd \rangle$ and the $\langle d^\dagger d^\dagger \rangle$ channels. Hence we can write

$$H_{exc} = J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle d^\dagger_{m'\downarrow} d_{m\downarrow} + \langle d^\dagger_{m'\downarrow} d_{m\downarrow} \rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m\downarrow} \rangle d^\dagger_{m'\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m'\downarrow} d_{m'\uparrow} \rangle d^\dagger_{m\uparrow} d_{m\downarrow} \right] \tag{25}$$

with a correction to the energy shift given by the following

$$\Delta E_{exc} = J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle \langle d^\dagger_{m'\downarrow} d_{m\downarrow} \rangle - \langle d^\dagger_{m\uparrow} d_{m\downarrow} \rangle \langle d^\dagger_{m'\downarrow} d_{m'\uparrow} \rangle \right] \tag{26}$$

Likewise we can write

$$H_{ph} = J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle d^\dagger_{m\downarrow} d_{m'\downarrow} + \langle d^\dagger_{m\downarrow} d_{m'\downarrow} \rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m'\downarrow} \rangle d^\dagger_{m\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m\downarrow} d_{m'\uparrow} \rangle d^\dagger_{m\uparrow} d_{m'\downarrow} \right] \tag{27}$$

$$\Delta E_{ph} = J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle \langle d^\dagger_{m\downarrow} d_{m'\downarrow} \rangle - \langle d^\dagger_{m\uparrow} d_{m'\downarrow} \rangle \langle d^\dagger_{m\downarrow} d_{m'\uparrow} \rangle \right] \tag{28}$$

7 Complete Kanamori Hamiltonian

$$\begin{aligned} H^k_{MF} = {} & U \sum_m \langle n_{m\uparrow} \rangle \hat{n}_{m\downarrow} + \langle n_{m\downarrow} \rangle \hat{n}_{m\uparrow} \qquad (29) \\ & + U' \sum_{m \neq m'} \langle n_{m\uparrow} \rangle \hat{n}_{m'\downarrow} + \langle n_{m'\downarrow} \rangle \hat{n}_{m\uparrow} \\ & + (U' - J) \sum_{m < m', \sigma} \langle n_{m\sigma} \rangle \hat{n}_{m'\sigma} + \langle n_{m'\sigma} \rangle \hat{n}_{m\sigma} \\ & + J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle d^\dagger_{m'\downarrow} d_{m\downarrow} + \langle d^\dagger_{m'\downarrow} d_{m\downarrow} \rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m\downarrow} \rangle d^\dagger_{m'\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m'\downarrow} d_{m'\uparrow} \rangle d^\dagger_{m\uparrow} d_{m\downarrow} \right] \\ & + J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle d^\dagger_{m\downarrow} d_{m'\downarrow} + \langle d^\dagger_{m\downarrow} d_{m'\downarrow} \rangle d^\dagger_{m\uparrow} d_{m'\uparrow} - \langle d^\dagger_{m\uparrow} d_{m'\downarrow} \rangle d^\dagger_{m\downarrow} d_{m'\uparrow} - \langle d^\dagger_{m\downarrow} d_{m'\uparrow} \rangle d^\dagger_{m\uparrow} d_{m'\downarrow} \right] \end{aligned}$$

Where we shift the energy by

$$\begin{aligned} \Delta E = {} & -U \sum_m \langle n_{m\uparrow} \rangle \langle n_{m\downarrow} \rangle - U' \sum_{m \neq m'} \langle n_{m\uparrow} \rangle \langle n_{m'\downarrow} \rangle \qquad (30) \\ & - (U' - J) \sum_{m < m', \sigma} \langle n_{m\sigma} \rangle \langle n_{m'\sigma} \rangle \\ & + J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle \langle d^\dagger_{m'\downarrow} d_{m\downarrow} \rangle - \langle d^\dagger_{m\uparrow} d_{m\downarrow} \rangle \langle d^\dagger_{m'\downarrow} d_{m'\uparrow} \rangle \right] \\ & + J \sum_{m \neq m'} \left[\langle d^\dagger_{m\uparrow} d_{m'\uparrow} \rangle \langle d^\dagger_{m\downarrow} d_{m'\downarrow} \rangle - \langle d^\dagger_{m\uparrow} d_{m'\downarrow} \rangle \langle d^\dagger_{m\downarrow} d_{m'\uparrow} \rangle \right] \end{aligned}$$

8 Density matrix

The density matrix for the grand canonical ensemble is

$$\hat{\rho} = \frac{e^{-\beta(\hat{H} - \mu \hat{N})}}{Z} \tag{31}$$

with $Z = \text{Tr } e^{-\beta(H-\mu N)}$. The Hamiltonian is non-interacting so it can be written on the energy basis

$$\hat{H} = \sum_n \varepsilon_n d_n^\dagger d_n \tag{32}$$

where the creation operator in this basis is defined by $d_n^\dagger = \sum_k \langle k|n\rangle c_k^\dagger$. Our fock space is defined by the occupation number in each energy state, for our fermion system this is either zero or one. We label the Fock state like $|\{n_k\}\rangle$. These are eigenstates of the Hamiltonian

$$\hat{H}|\{n_k\}\rangle = \sum_j \varepsilon_j \hat{d}_j^\dagger \hat{d}_j \left(\prod_k \left(d_k^\dagger\right)^{n_k}\right) \tag{33}$$

For $j \neq k$ the commutation is zero and we can flip the two to normal order which results in zero. The only non zero is for when $j = k$. Therefore for a single fock state we can write

$$\hat{H}|\{n_k\}\rangle = \sum_j \varepsilon_j n_j |\{n_k\}\rangle \tag{34}$$

the same reasoning allows us to write

$$\hat{N}|\{n_k\}\rangle = \sum_j n_j |\{n_k\}\rangle \tag{35}$$

using these results we find the partition function

$$\begin{aligned} Z &= \sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}|\{n_k\}\rangle \\ &= \sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\sum_j \beta(\varepsilon_j-\mu)n_j}|\{n_k\}\rangle \\ &= \sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\prod_j\langle\{n_k\}|e^{-\beta(\varepsilon_j-\mu)n_j}|\{n_k\}\rangle \end{aligned} \tag{36}$$

The sums out front go from 0, 1 since these are fermions. This means that the sums decouple and we can bring the product outside. Then we evaluate one sum from 0 to 1 and we get

$$Z = \prod_j \left(1 + e^{-\beta(\varepsilon_j-\mu)}\right) \tag{37}$$

Next we are interested in the density

$$\begin{aligned} \langle d_m^\dagger d_n\rangle &= \frac{\text{Tr}\left(e^{-\beta(\hat{H}-\mu\hat{N})}\hat{d}_m^\dagger d_n\right)}{Z} \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}\hat{d}_m^\dagger d_n|\{n_k\}\rangle \end{aligned} \tag{38}$$

Since our $\hat{H}$ and $\hat{N}$ are diagonal the density matrix is diagonal in this basis and $m = n$.

$$\begin{aligned} \langle d_n^\dagger d_n\rangle &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}\hat{d}_n^\dagger d_n|\{n_k\}\rangle \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\beta(\hat{H}-\mu\hat{N})}n_n|\{n_k\}\rangle \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|e^{-\sum_j \beta(\varepsilon_j-\mu)n_j}n_n|\{n_k\}\rangle \\ &= \frac{1}{Z}\sum_{n_0}\sum_{n_1}\cdots\sum_{n_N}\langle\{n_k\}|n_n\prod_j e^{-\beta(\varepsilon_j-\mu)n_j}|\{n_k\}\rangle \end{aligned} \tag{39}$$

We can cancel out all $j \neq n$ terms from the partition function in the denominator. Hence we will get

$$\begin{aligned} \langle d_n^\dagger d_n \rangle &= \frac{\sum_{n=0}^{1} n_n e^{-\beta(\varepsilon_n - \mu)n_n}}{1 + e^{-\beta(\varepsilon_n - \mu)}} \\ &= \frac{e^{-\beta(\varepsilon_n - \mu)}}{1 + e^{-\beta(\varepsilon_n - \mu)}} \\ &= f(\varepsilon_n - \mu) \end{aligned} \tag{40}$$

where f is the fermi dirac distribution. Now we perform a change of basis to the t2g basis. Recall $d_n^\dagger = \sum_k \langle k|n\rangle c_k^\dagger$. From the definition of the density matrix we can write

$$\begin{aligned} \rho_{ab} = \langle c_a^\dagger c_b \rangle &= \langle \left(\sum_k \langle a|k\rangle d_k^\dagger \right) \left(\sum_q \langle q|b\rangle d_q \right) \rangle \\ &= \sum_{kq} \langle a|k\rangle \langle q|b\rangle \langle d_k^\dagger d_q \rangle \\ &= \sum_{kq} \langle a|l\rangle \langle q|b\rangle f(\varepsilon_k - \mu)\delta_{kq} \\ &= \sum_n \langle a|n\rangle \langle n|b\rangle f(\varepsilon_k - \mu) \end{aligned} \tag{41}$$

Where $|n\rangle$ labels the eigenstates and a, b labels the t2g,spin band.